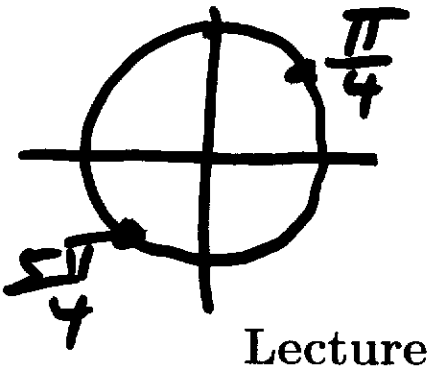
F12 4-1 AR

Lecture 33: Section 5.2
Verifying Trigonometric Identities

Verifying trigonometric identities

Conditional equation vs an identity



$x = \frac{\pi}{4} + n\pi$ where n is an integer.

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$x+2=5 \quad x=3$

Recall: A **conditional equation** is an equation that is true for only some of the values in its domain.

$2x+6 = 2x+6$ all reals

ex. $\tan x = 1$

$2x+6 = 2(x+3)$

An **identity** is an equation that is true for all real values in its domain.

$\sin^2 \frac{\pi}{3} + \cos^2 \frac{\pi}{12} = 1$

ex. $\sin^2 x = 1 - \cos^2 x$

$\sin^2 x + \cos^2 x = 1 \qquad x = \text{all reals}$

Guidelinesfor Verifying Trigonometric Identities Manipulate until both sides are the same.

1. Start with one side. It's usually easier to start with the more complicated side.

2. Use known identities. Bring fractional expressions to a common denominator, factor, and use the fundamental identities to simplify expressions.

3. Convert to sines and cosines. Sometimes it is helpful to rewrite all functions in terms of sines and cosines.

LHS

mult by the conjugate

$\tan^2\theta + 1 = \sec^2\theta$

<u>ex.</u> Verify the trigonometric identities.

1) $\dfrac{1}{\sec x - \tan x} = \sec x + \tan x$

$$LHS = \left(\frac{1}{\sec x - \tan x}\right)\left(\frac{\sec x + \tan x}{\sec x + \tan x}\right) = \frac{\sec x + \tan x}{\sec^2 x - \tan^2 x} \leftarrow 1$$

$$LHS = \frac{\sec x + \tan x}{1} = \sec x + \tan x = RHS \checkmark$$

$\sin^2\theta + \cos^2\theta = 1$

2) $\dfrac{\sin^2\theta}{\cos\theta} = \sec\theta - \cos\theta$

$$RHS = \sec\theta - \cos\theta = \frac{1}{\cos\theta} - \cos\theta\left(\frac{\cos\theta}{\cos\theta}\right)$$

$$RHS = \frac{\overbrace{1-\cos^2\theta}^{=\sin^2\theta}}{\cos\theta} = \frac{\sin^2\theta}{\cos\theta} = LHS \checkmark$$

Add together.

ex. Verify the trigonometric identity

$$2\tan x \sec x = \frac{1}{1-\sin x} - \frac{1}{1+\sin x}$$

$$RHS = \left(\frac{1}{1-\sin x}\right)\left(\frac{1+\sin x}{1+\sin x}\right) - \left(\frac{1}{1+\sin x}\right)\left(\frac{1-\sin x}{1-\sin x}\right) = \frac{1+\sin x - (1-\sin x)}{1-\sin^2 x} =$$

$$RHS = \frac{\cancel{1}+\sin x \cancel{-1} + \sin x}{\underbrace{1-\sin^2 x}_{=\cos^2 x}} = \frac{2\sin x}{\cos^2 x} = 2\left(\frac{\sin x}{\cos x}\right)\left(\frac{1}{\cos x}\right)$$

$$RHS = 2\tan x \sec x = LHS \checkmark$$

x LHS by its conjugate

<u>ex.</u> Verify the trigonometric identity

$$\frac{\cos\theta}{1-\sin\theta} = \sec\theta + \tan\theta$$

$$LHS = \left(\frac{\cos\theta}{1-\sin\theta}\right)\left(\frac{1+\sin\theta}{1+\sin\theta}\right) = \frac{\cos\theta(1+\sin\theta)}{1-\sin^2\theta} =$$

$$LHS = \frac{\cos\theta(1+\sin\theta)}{\cos^2\theta} = \frac{1+\sin\theta}{\cos\theta} = \overset{\sec\theta}{\frac{1}{\cos\theta}} + \overset{\tan\theta}{\frac{\sin\theta}{\cos\theta}} =$$

$$LHS = \sec\theta + \tan\theta = RHS \checkmark$$

Add LHS together

<u>ex.</u> Verify the trigonometric identity

$$\frac{\sin\theta}{1+\cos\theta} + \frac{1+\cos\theta}{\sin\theta} = 2\csc\theta$$

$$LHS = \left(\frac{\sin\theta}{1+\cos\theta}\right)\left(\frac{\sin\theta}{\sin\theta}\right) + \left(\frac{1+\cos\theta}{\sin\theta}\right)\left(\frac{1+\cos\theta}{1+\cos\theta}\right) =$$

$$LHS = \frac{\sin^2\theta + 1 + 2\cos\theta + \cos^2\theta}{\sin\theta(1+\cos\theta)} = \frac{2+2\cos\theta}{\sin\theta(1+\cos\theta)}$$

$$LHS = \frac{2\cancel{(1+\cos\theta)}}{\sin\theta\cancel{(1+\cos\theta)}} = \frac{2}{\sin\theta} = 2\csc\theta = RHS \checkmark$$

$$2 \cdot \frac{1}{\sin\theta}$$

LHS change to sin/cos

<u>ex.</u> Verify the trigonometric identity

$$\frac{\tan\theta + \cot\theta}{\sec\theta\csc\theta} = 1.$$

$$LHS = \frac{\frac{\sin\theta}{\cos\theta}\left(\frac{\sin\theta}{\sin\theta}\right) + \frac{\cos\theta}{\sin\theta}\left(\frac{\cos\theta}{\cos\theta}\right)}{\frac{1}{\cos\theta}\cdot\frac{1}{\sin\theta}} = \frac{\frac{\sin^2\theta + \cos^2\theta}{\sin\theta\cos\theta}}{\frac{1}{\cos\theta\sin\theta}} = \quad = 1$$

$$LHS = \frac{\frac{1}{\sin\theta\cos\theta}}{\frac{1}{\sin\theta\cos\theta}} = 1 = RHS \checkmark$$

Sometimes, it is practical to work with each side separately to obtain one common form equivalent to both sides.

<u>ex.</u> Verify the trigonometric identity $\times$ by the conjugate

split up

$$\frac{1+\cos\theta}{\cos\theta} = \frac{\tan^2\theta}{\sec\theta - 1}$$

$$LHS = \frac{1+\cos\theta}{\cos\theta} = \frac{1}{\cos\theta} + \frac{\cos\theta}{\cos\theta} = \sec\theta + 1$$

$$RHS = \left(\frac{\tan^2\theta}{\sec\theta - 1}\right)\left(\frac{\sec\theta+1}{\sec\theta+1}\right) = \frac{\tan^2\theta(\sec\theta+1)}{\underbrace{\sec^2\theta - 1}_{=\tan^2\theta}}$$

$$RHS = \frac{\cancel{\tan^2\theta}(\sec\theta+1)}{\cancel{\tan^2\theta}} = \sec\theta + 1 = LHS \checkmark$$

Split up sin x on LHS, since there is a sin x on RHS.

Calculus Examples

<u>ex.</u> Verify:

1) $\sin^3 x \cos^2 x = (\cos^2 x - \cos^4 x)\sin x$

$LHS = \sin x(\sin^2 x \cos^2 x)$

$LHS = \sin x\,(1-\cos^2 x)\cos^2 x = \sin x(\cos^2 x - \cos^4 x) = RHS$ ✓

$(\sec^2 x)^2 = ((\sec x)^2)^2$

$(\sec x \cdot \sec x)^2$

on LHS pull out $\sec^2 x$

2) $\sec^6 x \tan x = (\tan x + 2\tan^3 x + \tan^5 x)\sec^2 x$

$LHS = \sec^2 x(\sec^4 x \tan x) =$

$LHS = \sec^2 x\,(\sec^2 x)^2 \tan x =$

$LHS = \sec^2 x\,(\tan^2 x + 1)^2 \tan x$

$LHS = \sec^2 x(\tan^4 x + 2\tan^2 x + 1)\tan x$

$LHS = \sec^2 x(\tan^5 x + 2\tan^3 x + \tan x) = RHS$ ✓